

HOMEWORK ASSIGNMENT 5

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

QUESTION 1: Commute Times.

The United States Census Bureau routinely publishes data (including predictive data) on their website. According to one of their published sources¹, the 2024 average (mean) commute time among all U.S. residents ages 16 or over (specifically, among those who did not work from home) was 27.2 minutes.

Suppose Eric wants to determine whether or not the average commute time among San Luis Obispo residents in 2024 is less than the national average (as reported by the U.S. Census Bureau). To that end, he obtained a simple random sample of 100 San Luis Obispo commuters (back in 2024); shown below are some summary statistics of Eric's data:

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 26.5$ min.	median = 23.0 min.	$s = 4.805$ min.	$n = 59$

Table 1: Summary statistics of Eric's data.

- (a) Describe one observational unit, in the context of this study.
- (b) Define the parameter in words, and identify its appropriate mathematical symbol.
- (c) Write down the null and alternative hypotheses using symbols and math only.
- (d) Can we use a theory-based approach to test these hypotheses? Why or why not? **Hint:** is there any information missing?

Use the following for the remaining parts of this problem: We will proceed using a simulation-based approach. The plot below shows a dotplot representing a simulated null distribution, where I used **150** simulated samples (note that I did not run our usual 10,000 repetitions!)

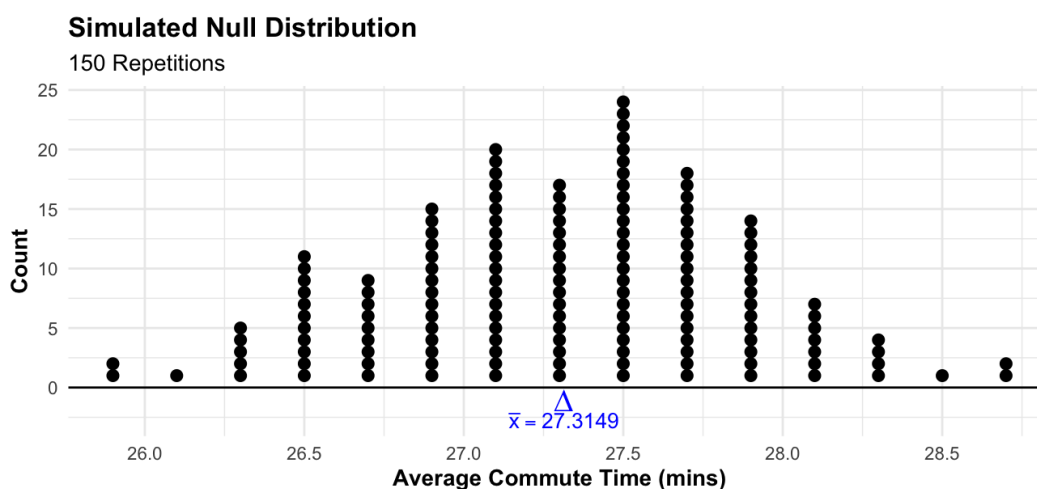


Figure 1: Simulated null distribution.

¹ <https://www2.census.gov/programs-surveys/commuting/guidance/acs-1yr/Mean-travel-time.pdf>

The mean of this simulated null distribution is 27.32 and the standard deviation (also of this simulated null distribution) is 0.55.

- (e) Calculate and report the p -value. Show your work!
- (f) Calculate the simulation-based t -statistic.
- (g) Calculate the theory-based t -statistic (regardless of whether you think a theory-based approach is valid here or not).
- (h) How justified are we in generalizing the results of this study to the population of interest? In other words, describe the potential scope of inference of this study based on the sampling method described.
- (i) Answer the research question. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p -value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.
- (j) Eric's friend, Erika, is interested not in whether the average commute time among San Luis Obispo residents is less than the national average (of 27.2 minutes), but merely whether the average commute time among San Luis Obispo residents *differs* from the national average. If Erika uses the same data that Eric did, what would be her approximate p -value? Do **not** use any applets for this; also, as a hint, you shouldn't need to re-count anything (though you may need to recalculate certain quantities).

QUESTION 2: Reaching New Heights.

Based on statistics from the [Centers for Disease Control](#) (CDC), we can infer that they claim the average height of a US resident is 66.2 inches.² Suppose I (Dr. Ethan) think that's a little low. In particular, I'd like to address the question: is the true average height of a US resident larger than 66.2 inches?

As a part of the Welcome Survey I sent out at the start of the semester, I asked all of you (all three of my STAT 1110 lecture sections) what your height is in inches, without shoes. After cleaning up responses (converting feet to inches, removing obvious mis-inputs, etc.) I was left with a total of 82 responses. Shown below is a dotplot of these 82 height measurements, along with some summary statistics.

²Technically they only report an average height for "men" and "women."

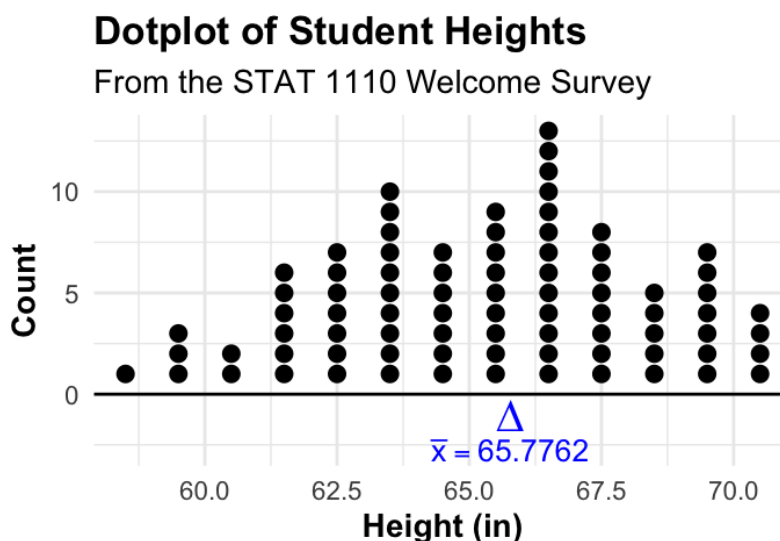


Figure 2: Results from the Welcome Survey.

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 65.776$ in.	median = 66 in.	$s = 2.999$ in.	$n = 82$

Table 2: Summary statistics of the data from the Welcome Survey.

- What does one dot on Figure (2) represent?
- Define the parameter of interest, and identify its appropriate mathematical symbol.
- Write down the null and alternative hypotheses using proper mathematical symbols and numbers.
- Are the validity conditions met if we want to conduct an analysis based on the theory-based approach (i.e. a one-sample t -test)? Explain.

Regardless of your answer to the previous part, we'll proceed with a theory-based approach for the remainder of this problem.

- Calculate the value of the t -statistic.
- Navigate to the [Theory-Based Inference Applet](#) and calculate the theory-based p -value.
- Evaluate** and **Interpret** the p -value. (At this point, I will rely on you to remember what these mean!)
- How justified are we in generalizing the results of this study to the population of interest? In other words, describe the potential scope of inference of this study based on the sampling method described.
- Answer the research question. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p -value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) **reject/fail to reject the null hypothesis**, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.

QUESTION 3: Notation Check-In

In each of the following situations, you will be presented with several quantities. Identify the correct mathematical symbol for the indicated quantity.

- (a) Pia wants to determine whether the the proportion of households in a rural county that live under the poverty line exceeds the national average. She took a sample of 12 households, and found that 8 of them fall under the poverty line. **What symbol should be used to denote the quantity 8/12?**
- (b) Katrina wants to determine the average weight across all hummingbirds in California. A convenience sample of 50 hummingbirds had an average weight of 5.34 grams.
 - i. What symbol should be used to denote the quantity 5.34 grams?
 - ii. What symbol should be used to denote the quantity 50?
- (c) Sneha has read an article that claims the average height of all palm trees in the Brazilian Amazon is 57.41 feet. She takes random sample of 64 palm trees, and finds this sample of trees to have an average height of 54.32 feet and a standard deviation of 12.43 feet. Unbeknownst to Sneha or the authors of the article, the set of all palm trees in the Brazilian Amazon have an average height of 59.23 feet and a standard deviation of 17.44 feet.
 - i. What symbol should be used to denote the quantity 57.41 feet?
 - ii. What symbol should be used to denote the quantity 54.32 feet?
 - iii. What symbol should be used to denote the quantity 12.43 feet?
 - iv. What symbol should be used to denote the quantity 59.23 feet?
 - v. What symbol should be used to denote the quantity 17.44 feet?
 - vi. What symbol should be used to denote the quantity 64 feet?

QUESTION 4: Type I and Type II Errors

As with a jury trial, another analogy to hypothesis testing involves medical diagnostic tests. These tests aim to indicate whether or not the patient has a particular disease. But the tests are not infallible, so errors can be made. The null hypothesis can be regarded as the patient being healthy (the baseline). The alternative hypothesis can be regarded as the patient having the disease.

- (a) Describe what Type I error (false alarm) represents in this situation.
 - ☐ The patient does not have the disease, but the test says the patient has the disease
 - ☐ The patient has the disease, but the test says the patient is healthy.
- (b) Describe what Type II error (missed opportunity) represents in this situation.
 - ☐ The patient does not have the disease, but the test says the patient has the disease
 - ☐ The patient has the disease, but the test says the patient is healthy.